Can Atmospheric Refraction Reproduce Geometric Horizon Occlusion?

July 31, 2026

Abstract

We investigate whether vertical atmospheric refraction above a flat, opaque boundary can reproduce the horizon occlusion normally associated with a spherical surface. Starting from Fermat's principle, we derive the ray equation for a stratified refractive index and show that, in the paraxial limit, an exponential profile reproduces the leading-order spherical horizon law. The exponential model is also solved exactly and shown, through a conformal coordinate transformation, to be optically equivalent to propagation outside a circle. We compare its exact horizon law with that of a sphere and derive a general inverse formula for reconstructing the refractive-index profile associated with any prescribed grazing-distance law. The resulting flat-surface constructions require the refractive index to increase with height, whereas ordinary atmospheric refractivity decreases with height. Thus refraction can be engineered to imitate geometric horizon occlusion in a mathematical model, but the required profile is not characteristic of the normal atmosphere.

1 Introduction

The bottom-first disappearance of distant objects behind a horizon is normally attributed to geometric occlusion by a curved surface. Atmospheric refraction and its role in astronomical and terrestrial observations have a long history [7, 10]. A vertically varying refractive index offers a conceptually different mechanism: it bends light rays and may prevent light from the lower part of a distant object from reaching an observer even when the underlying boundary is flat. The observation that refraction bends light is not, by itself, sufficient to show that it can reproduce the visibility pattern of a spherical surface. Demonstrating such a mechanism requires an explicit refractive-index profile together with the corresponding family of ray trajectories.

We formulate this question as an inverse problem in geometrical optics, using Fermat's principle and the standard ray description of a stratified medium [2, 10]. Consider a flat, opaque boundary covered by a smooth, vertically stratified medium with refractive index $n(h)$. We ask whether this system can reproduce the principal horizon effects associated with a sphere of radius R: bottom-first disappearance, recovery with observer elevation, and a limiting visible distance determined by observer and target height.

The problem naturally separates into mathematical and physical components. The mathematical question is whether a profile $n(h)$ producing the required ray geometry exists. The physical question is whether the sign and magnitude of such a profile are compatible with an ordinary atmosphere. Mathematical existence is therefore not taken as evidence that the corresponding mechanism occurs under terrestrial conditions.

The remainder of the paper develops this program. In the paraxial limit, an exponential profile that increases with height produces upward-curving rays and recovers the leading-order spherical horizon law. The same profile also admits an exact ray solution. A conformal change of variables maps its two-dimensional optical metric to the Euclidean exterior of a circle, thereby explaining why a flat boundary can acquire the qualitative occlusion geometry of a curved one. Differential-geometric formulations of atmospheric ray propagation provide a broader context for this optical-metric viewpoint [6]. We then compare the exact refractive and spherical horizon laws and derive a general inverse formula that reconstructs a stratified refractive index from a prescribed horizon-distance law.

The physical comparison provides the decisive distinction. The globe-like construction requires $dn/dh > 0$: the refractive index must increase with altitude rather than decrease, contrary to the behavior of the ordinary atmosphere, for which $dn/dh < 0$. The desired optical occlusion is therefore mathematically realizable above a flat surface, but only in a medium whose vertical trend is opposite to that of the normal atmosphere.

1.1 Scope and assumptions

The analysis uses geometrical optics and assumes a stationary, isotropic, lossless medium whose refractive index depends only on height. The lower boundary is flat and perfectly opaque. The profile $n(h)$ is smooth throughout the region traversed by the rays, and propagation is reciprocal. Diffraction, scattering, absorption, turbulence, and time-dependent atmospheric structure are excluded. These assumptions isolate the question of whether vertical refraction alone can reproduce horizon-like occlusion; they do not constitute a complete model of atmospheric imaging.

The construction is developed in two dimensions because every ray in a vertically stratified medium lies in a vertical plane. The analysis concerns the limiting ray geometry and visibility threshold; full three-dimensional image formation, transverse magnification, and distortion lie outside its scope. Throughout, R denotes the radius of the spherical surface whose horizon geometry serves as the comparison target; it is not an assumption about the geometry of the flat optical model.

2 Stratified Optical Model

2.1 Geometry and assumptions

We consider a two-dimensional vertical section through a horizontally homogeneous medium. Let x denote horizontal distance and let $h \geq 0$ denote height above an opaque planar boundary located at $h = 0$. The refractive index is assumed to be isotropic and vertically stratified,

$$n = n(h), \qquad n(h) > 0, \tag{1}$$

with

$$n_0 \equiv n(0). \tag{2}$$

Throughout the paper, h is measured upward from the opaque boundary. Horizontal translational symmetry allows each ray to be analysed in a single vertical plane. A ray is represented by a curve $h = h(x)$ wherever it is single-valued in x.

The boundary at $h = 0$ is assumed to be perfectly opaque. A source and an observer are mutually visible only if at least one geometrical-optics ray connecting them remains in the open half-space $h > 0$, except possibly at a single tangency point in the limiting case.

2.2 Fermat's principle and the first integral

The optical path length of a ray between two fixed endpoints is, in the geometrical-optics approximation [2],

$$\mathcal{S}[h] = \int n(h)\, ds = \int n(h)\sqrt{1 + h'^2}\, dx, \tag{3}$$

where $h' = dh/dx$. Geometrical-optics rays are stationary paths of the functional in Eq. (3). The corresponding Lagrangian is

$$\mathcal{L}(h, h') = n(h)\sqrt{1 + h'^2}. \tag{4}$$

It contains no explicit dependence on x. The associated horizontal translational symmetry therefore supplies a first integral: the Beltrami identity reduces the Euler–Lagrange equation to

$$\mathcal{L} - h'\frac{\partial \mathcal{L}}{\partial h'} = \frac{n(h)}{\sqrt{1 + h'^2}} = a, \tag{5}$$

where a is constant along a ray. Introducing the angle θ between ray and horizontal, so that $\cos\theta = 1/\sqrt{1 + h'^2}$, Eq. (5) becomes

$$n(h)\cos\theta = a. \tag{6}$$

This is the continuous form of Snell's law appropriate to a horizontally stratified medium [10, 8].

Rearranging Eq. (5) gives the ray slope in the form

$$h'^2 = \frac{n^2(h)}{a^2} - 1. \tag{7}$$

On any monotonic branch, inverting this relation gives

$$\left|\frac{dx}{dh}\right| = \frac{a}{\sqrt{n^2(h) - a^2}}. \tag{8}$$

These relations support both the direct construction of ray trajectories and the inverse reconstruction of a refractive-index profile from a prescribed visibility law.

2.3 Grazing rays and horizon distance

For fixed endpoint heights, the limiting visible ray just grazes the opaque boundary. At its point of contact with $h = 0$ the ray is horizontal, so that $h' = 0$. Equation (5) then fixes its constant to

$$a = n_0. \tag{9}$$

Such a grazing ray requires $n(h) \geq n_0$ throughout the heights it traverses. Hence the upward-bending profiles considered here satisfy $n(h) \geq n_0$ near the boundary.

Define $d(h)$ as the maximum horizontal distance from a boundary tangency point to an endpoint at height h. Using Eq. (9) in Eq. (8) gives

$$d(h) = \int_0^h \frac{n_0\, d\eta}{\sqrt{n^2(\eta) - n_0^2}}. \tag{10}$$

The square-root singularity at the lower limit is integrable whenever the profile rises regularly from n_0.

Let an observer and a target have heights h_o and h_t, respectively. By horizontal translational symmetry, the grazing point may be chosen as the origin. The limiting ray then consists of two reciprocal branches meeting tangentially at the boundary, and the maximum horizontal separation is

$$L_{\max}(h_\mathrm{o}, h_\mathrm{t}) = d(h_\mathrm{o}) + d(h_\mathrm{t}). \tag{11}$$

Figure 1 summarizes the corresponding one-sided horizon distance and the two-sided limiting geometry. For separations below this limit, an unobstructed ray exists. At equality, the limiting ray grazes the boundary. Beyond it, every connecting ray intersects the opaque surface. Equation (10) therefore provides the basic link between a stratified refractive-index profile and the resulting horizon-occlusion law.

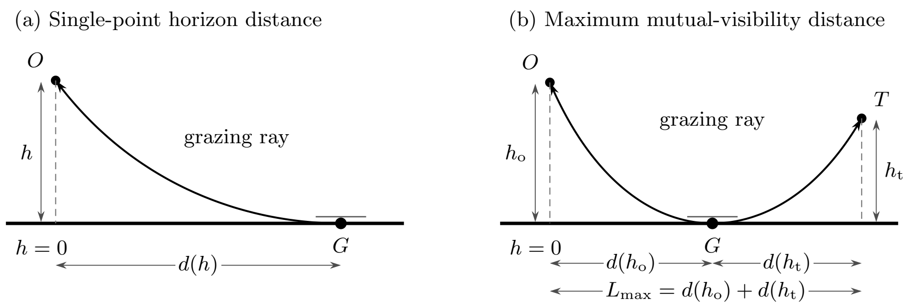

Figure 1: Grazing-ray geometry above an opaque planar boundary. (a) The horizon distance $d(h)$ from a boundary tangency point G to an endpoint O at height h. (b) The limiting mutual-visibility ray between an observer at height h_o and a target at height h_t; its two reciprocal branches meet with a horizontal tangent at G, giving $L_\mathrm{max} = d(h_\mathrm{o}) + d(h_\mathrm{t})$.

3 Paraxial Refractive Solution

We now seek the refractive-index profile that reproduces the leading-order horizon geometry within the paraxial approximation.

3.1 Small-angle ray equation

We first restrict attention to rays with small slope relative to the horizontal, as in standard paraxial treatments of stratified atmospheric refraction [8],

$$|h'| \ll 1. \tag{12}$$

Expanding the Fermat integrand in Eq. (3) gives

$$n(h)\sqrt{1+h'^2} = n(h)\left(1 + \frac{h'^2}{2}\right) + \mathcal{O}(h'^4). \tag{13}$$

The Euler–Lagrange equation for the retained terms in Eq. (13) is

$$\frac{d}{dx}\big(n(h)h'\big) - \frac{dn}{dh}\left(1 + \frac{h'^2}{2}\right) = 0. \tag{14}$$

Neglecting terms of order h'^2 then yields

$$n(h)h'' \simeq \frac{dn}{dh}, \tag{15}$$

or, equivalently,

$$h'' \simeq \frac{d\ln n}{dh}. \tag{16}$$

Thus, within the paraxial approximation, the logarithmic vertical gradient of the refractive index determines the local ray curvature.

3.2 Profile producing constant ray curvature

Relative to a local horizontal tangent plane, the leading-order horizon geometry of a sphere of radius R is generated by curves with constant upward curvature $1/R$. To reproduce that ray family above a flat boundary, we therefore impose

$$h'' = \kappa, \qquad \kappa > 0, \tag{17}$$

with $\kappa = 1/R$ for the spherical comparison. Substituting Eq. (17) into Eq. (16) gives

$$\frac{d \ln n}{dh} = \kappa. \tag{18}$$

Integrating with the boundary condition $n(0) = n_0$ yields

$$n(h) = n_0 e^{\kappa h}. \tag{19}$$

For the spherical comparison scale,

$$\kappa = \frac{1}{R}, \tag{20}$$

so that

$$n(h) = n_0 e^{h/R}. \tag{21}$$

The exponential form is therefore derived from the constant-curvature requirement rather than assumed. The required refractive index increases with height, a sign that will be central to the later comparison with an ordinary atmosphere.

3.3 Parabolic rays and the visibility law

The corresponding ray family follows by integrating Eq. (17):

$$h(x) = h_m + \frac{\kappa}{2}(x - x_m)^2, \tag{22}$$

where (x_m, h_m) is the vertex of the parabola. A limiting ray that just grazes the opaque boundary has $h_m = 0$, and its tangency point may be chosen as $x_m = 0$. Its two branches are therefore

$$h(x) = \frac{\kappa x^2}{2}. \tag{23}$$

An endpoint at height h lies on this grazing ray at horizontal distance

$$d_{\mathrm{p}}(h) = \sqrt{\frac{2h}{\kappa}}. \tag{24}$$

Consequently, for observer and target heights h_{o} and h_{t}, the maximum separation is

$$L_{\max}^{(\mathrm{p})} = \sqrt{\frac{2h_{\mathrm{o}}}{\kappa}} + \sqrt{\frac{2h_{\mathrm{t}}}{\kappa}}. \tag{25}$$

Setting $\kappa = 1/R$ gives

$$L_{\max}^{(\mathrm{p})} = \sqrt{2Rh_{\mathrm{o}}} + \sqrt{2Rh_{\mathrm{t}}}. \tag{26}$$

This is precisely the leading-order spherical horizon law for heights much smaller than R.

The paraxial construction therefore establishes that a flat-boundary model can reproduce the leading spherical horizon-occlusion law: the lower parts of a distant object become inaccessible first, and raising either endpoint increases the limiting visible distance according to the same square-root height dependence as on a sphere. The agreement is asymptotic, however. The next section retains the same exponential refractive-index profile, drops the small-angle approximation, and solves the ray equations exactly. This determines which features of the paraxial result remain exact and which are specific to the asymptotic regime.

4 Exact Exponential Solution

The paraxial construction identifies the exponential profile

$$n(h) = n_0 e^{\kappa h} \tag{27}$$

as the natural candidate for reproducing the spherical horizon law above a flat boundary. The first integral derived in Sec. 2 also makes this profile exactly solvable, without invoking the small-slope approximation.

4.1 Exact ray trajectory

For a ray whose minimum height is h_m, the slope vanishes at $x = x_m$. Evaluating the first integral there gives

$$a = n(h_m) = n_0 e^{\kappa h_m}. \tag{28}$$

Substitution into the exact slope equation yields

$$\left(\frac{dh}{dx}\right)^2 = \frac{n^2(h)}{n^2(h_m)} - 1 = e^{2\kappa(h-h_m)} - 1. \tag{29}$$

On either branch of the ray,

$$|x - x_m| = \int_{h_m}^{h} \frac{d\eta}{\sqrt{e^{2\kappa(\eta-h_m)} - 1}}. \tag{30}$$

Evaluating the integral gives

$$|x - x_m| = \frac{1}{\kappa} \arccos\Big[e^{-\kappa(h-h_m)}\Big]. \tag{31}$$

Inverting Eq. (31) produces the exact ray family

$$h(x) = h_m - \frac{1}{\kappa} \ln\Big[\cos\big(\kappa(x - x_m)\big)\Big], \tag{32}$$

valid on the interval

$$|\kappa(x - x_m)| < \frac{\pi}{2}. \tag{33}$$

Near the minimum, expansion of the logarithm gives

$$h(x) = h_m + \frac{\kappa}{2}(x - x_m)^2 + \frac{\kappa^3}{12}(x - x_m)^4 + O\big(\kappa^5 (x - x_m)^6\big)\,, \tag{34}$$

which recovers the parabolic result of the previous section at leading order.

The exact curvature is not constant away from the minimum. Indeed, Eq. (32) implies

$$h'' = \kappa \sec^2\!\big(\kappa(x - x_m)\big) = \kappa\left(1 + h'^2\right), \tag{35}$$

which reduces to $h'' \simeq \kappa$ only in the paraxial regime.

4.2 Exact grazing distance

A limiting ray that just touches the opaque boundary has

$$h_m = 0. \tag{36}$$

The horizontal distance from the tangency point to an endpoint at height h is therefore

$$d_{\mathrm{e}}(h) = \frac{1}{\kappa}\arccos\!\big(e^{-\kappa h}\big)\,. \tag{37}$$

For observer and target heights h_{o} and h_{t}, the exact limiting separation is

$$L^{(\mathrm{e})}_{\max} = \frac{1}{\kappa}\left[\arccos\!\big(e^{-\kappa h_{\mathrm{o}}}\big) + \arccos\!\big(e^{-\kappa h_{\mathrm{t}}}\big)\right]. \tag{38}$$

With $\kappa = 1/R$, this becomes

$$L^{(\mathrm{e})}_{\max} = R\left[\arccos\!\left(e^{-h_{\mathrm{o}}/R}\right) + \arccos\!\left(e^{-h_{\mathrm{t}}/R}\right)\right]. \tag{39}$$

Thus the exponential profile produces a genuine boundary-occlusion threshold above a flat surface: beyond the limiting separation, every connecting ray intersects the boundary, whereas raising either endpoint restores a non-intersecting ray.

4.3 Paraxial limit

For $z = \kappa h \ll 1$,

$$\arccos\!\big(e^{-z}\big) = \sqrt{2z}\left(1 - \frac{z}{6} + \frac{z^2}{120} + O(z^3)\right). \tag{40}$$

Consequently,

$$d_{\mathrm{e}}(h) = \sqrt{\frac{2h}{\kappa}}\left(1 - \frac{\kappa h}{6} + \frac{(\kappa h)^2}{120} + O\big((\kappa h)^3\big)\right). \tag{41}$$

The leading term is precisely the paraxial distance in Eq. (24). The higher-order terms quantify where the exponential refractive construction departs from a constant- curvature approximation. The next section compares this exact optical law with the corresponding horizon law on a spherical surface.

5 Conformal Equivalence to Circular Geometry

The closed-form solution of Sec. 4 admits a simple geometrical interpretation. Optical metrics and differential-geometric methods have also been used to formulate atmospheric refraction [6]. For the Earth-scale choice $\kappa = 1/R$, the optical line element associated with the exponential profile is

$$d\ell_{\rm opt}^2 = n_0^2 e^{2h/R} \left(dx^2 + dh^2\right). \tag{42}$$

Introduce the coordinates

$$r = Re^{h/R}, \qquad \phi = \frac{x}{R}. \tag{43}$$

Since $dh = R\,dr/r$ and $dx = R\,d\phi$, substitution into Eq. (42) gives

$$d\ell_{\rm opt}^2 = n_0^2 \left(dr^2 + r^2 d\phi^2\right). \tag{44}$$

Apart from the constant factor n_0^2, this is the Euclidean metric in polar coordinates. The exponential refractive medium above a flat boundary is therefore optically equivalent to propagation in the homogeneous Euclidean exterior of a circle. Figure 2 summarizes this correspondence: the curved grazing ray and planar boundary in physical coordinates become a straight tangent line and circular boundary in the transformed coordinates.

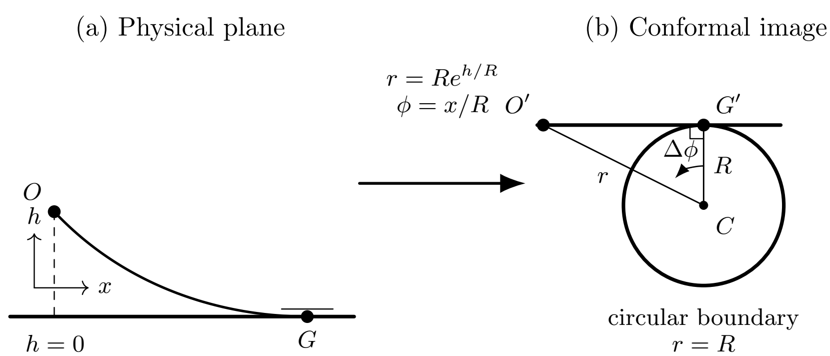

Figure 2: Conformal representation of the exponential refractive model. The transformation $r = Re^{h/R}$, $\phi = x/R$ maps the opaque planar boundary $h = 0$ to the circle $r = R$ and the curved grazing ray to a straight tangent line.

5.1 Boundary and ray mapping

The physical half-plane $h \geq 0$ is mapped to

$$r \geq R, \tag{45}$$

while the opaque flat boundary $h = 0$ becomes the circle $r = R$. Because multiplication of a metric by a positive constant does not alter its geodesics, rays in the transformed coordinates are ordinary straight lines. A limiting ray in the original medium maps to a line tangent to the circle. The apparent horizon generated by the refractive profile is therefore not merely a local analogy: it is the image under Eq. (43) of ordinary Euclidean occlusion by a circular boundary.

Let the tangency point correspond to $(x_m, h = 0)$, or equivalently to polar angle $\phi_m = x_m/R$. A line tangent to $r = R$ at that point satisfies

$$r \cos(\phi - \phi_m) = R. \tag{46}$$

Using Eq. (43), this becomes

$$e^{h/R} \cos\left(\frac{x - x_m}{R}\right) = 1, \tag{47}$$

and hence

$$h(x) = -R \ln\left[\cos\left(\frac{x - x_m}{R}\right)\right]. \tag{48}$$

Equation (48) is precisely the grazing member of the exact ray family in Eq. (32).

For a point at height h, the angular displacement between the point and the tangency radius follows directly from the right triangle formed by the circle centre, the tangency point, and the endpoint:

$$\Delta\phi = \arccos\left(\frac{R}{r}\right) = \arccos\left(e^{-h/R}\right). \tag{49}$$

Since $x = R\phi$, the corresponding physical horizontal distance is

$$d(h) = R \arccos\left(e^{-h/R}\right), \tag{50}$$

which recovers Eq. (37) for $\kappa = 1/R$ without further integration.

5.2 Meaning and limits of the equivalence

The equivalence is exact for the optical metric and its ray paths. It explains, in a single construction, why the exponential profile produces bottom-first disappearance, a limiting tangent ray, and recovery of hidden parts when the observer or target is raised. It also identifies these effects with boundary occlusion in the transformed geometry rather than with loss of angular resolution or a generic consequence of perspective.

The coordinate map does not, however, equate the physical height h above the flat boundary with the ordinary radial height above the circle. The latter is

$$r - R = R\left(e^{h/R} - 1\right) = h + \frac{h^2}{2R} + O\left(\frac{h^3}{R^2}\right). \tag{51}$$

Consequently, the refractive model is exactly equivalent to circular geometry after a nonlinear redefinition of height, while comparison at the same numerical physical height is only asymptotic for $h/R \ll 1$. The next section quantifies this distinction by comparing the exact refractive and spherical horizon laws.

6 Comparison with a Spherical Surface

The exponential refractive profile and a spherical surface produce the same leading square-root dependence of horizon distance on endpoint height, but they are not identical when the same numerical height is used in both models. We now compare the two exact laws and quantify their difference.

6.1 Exact spherical horizon law

Consider a sphere of radius R and a point at radial height h above its surface. Let α be the central angle between the point's radial line and the radius through the tangency point of the limiting line of sight. The corresponding right triangle gives

$$\cos\alpha = \frac{R}{R+h}. \tag{52}$$

The surface distance from the tangency point to the foot of the elevated point is therefore

$$d_{\mathrm{s}}(h) = R\alpha = R\arccos\left(\frac{R}{R+h}\right). \tag{53}$$

For observer and target heights h_{o} and h_{t}, the maximum surface separation before mutual occlusion is

$$L_{\max}^{(\mathrm{s})} = d_{\mathrm{s}}(h_{\mathrm{o}}) + d_{\mathrm{s}}(h_{\mathrm{t}}). \tag{54}$$

Here the comparison variable is surface arc length. This is the quantity corresponding to the horizontal coordinate $x = R\phi$ in the conformal map of Sec. 5. The straight-line distance from the elevated point to the tangency point,

$$\sqrt{h(2R+h)}, \tag{55}$$

is a different geometric quantity and is not used in the comparison below.

6.2 Exact refractive and spherical laws

For the Earth-scale exponential profile, Eq. (37) with $\kappa = 1/R$ gives

$$d_{\mathrm{e}}(h) = R \arccos\left(e^{-h/R}\right). \tag{56}$$

The two exact horizon laws are thus

$$d_{\mathrm{e}}(h) = R \arccos\left(e^{-h/R}\right), \tag{57}$$

$$d_{\mathrm{s}}(h) = R \arccos\left(\frac{1}{1 + h/R}\right). \tag{58}$$

Their difference arises entirely from the relation between physical height and transformed radial height. Under Eq. (43), a flat-model height h corresponds to the circular radius

$$r = Re^{h/R}. \tag{59}$$

Consequently, the exponential model is exactly equivalent to a circular model whose radial height is

$$H(h) = R\left(e^{h/R} - 1\right), \tag{60}$$

not to a circular model whose radial height is numerically equal to h. In fact,

$$d_{\mathrm{e}}(h) = d_{\mathrm{s}}(H(h)) \tag{61}$$

holds exactly. The mismatch at equal heights is therefore due to coordinate reparameterization, not a failure of the conformal equivalence.

6.3 Small-height comparison

Let

$$z = \frac{h}{R}. \tag{62}$$

For $z \ll 1$, the refractive distance has the expansion already obtained in Eq. (41),

$$d_{\mathrm{e}}(h) = \sqrt{2Rh}\left[1 - \frac{z}{6} + \frac{z^2}{120} + O(z^3)\right]. \tag{63}$$

The spherical arc distance expands as

$$d_{\mathrm{s}}(h) = \sqrt{2Rh}\left[1 - \frac{5z}{12} + \frac{43z^2}{160} + O(z^3)\right]. \tag{64}$$

Both laws therefore reduce to

$$d(h) \simeq \sqrt{2Rh} \tag{65}$$

at leading order. Their first difference appears one order later:

$$d_{\mathrm{e}}(h) - d_{\mathrm{s}}(h) = \sqrt{2Rh}\left[\frac{z}{4} - \frac{25z^2}{96} + O(z^3)\right]. \tag{66}$$

Hence the relative discrepancy is of order h/R. For ordinary terrestrial heights this is extremely small even though the two laws are not exactly the same at equal numerical height.

Figure 3 compares the two exact laws over a range large enough to expose their systematic separation while retaining the common small-height regime.

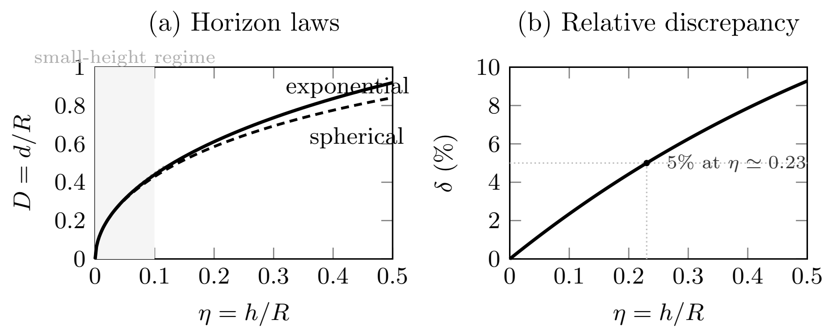

Figure 3: Comparison of the exponential-model and spherical horizon laws. Both reduce to the same small-height limit $D \sim \sqrt{2\eta}$, but diverge systematically as $\eta = h/R$ increases. Panel (b) shows the relative discrepancy.

6.4 Numerical scale of the discrepancy

Table 1 illustrates the difference for $R = 6371\,\text{km}$. The listed distances are one-sided horizon distances from a tangency point to an endpoint at height h.

h	d_e (km)	d_s (km)	$d_e - d_s$
2 m	5.048168	5.048167	0.40 mm
10 m	11.288044	11.288039	4.43 mm
100 m	35.695845	35.695705	0.140 m
1 km	112.877515	112.873086	4.43 m
10 km	356.866007	356.726164	139.84 m

Table 1: Exact exponential-refractive and spherical horizon distances at equal numerical height for $R = 6371\,\text{km}$.

At low altitude, the numerical agreement is substantially stronger than equality of the leading square-root term alone would suggest. Nevertheless, with sufficiently large heights or sufficiently precise measurements the two models are distinguishable. Exact agreement at equal physical height would require a different refractive-index profile, which can be reconstructed from the desired visibility law by the inverse method developed later in the paper.

7 General Inverse Reconstruction

The preceding sections began with a refractive-index profile and derived the resulting horizon law. The same relation can instead be read in reverse. Inverse methods have previously been applied to atmospheric-refraction data [9]; the reconstruction below is independently derived for the specific grazing-distance relation of the present model. Given a desired one-sided visibility distance as a function of endpoint height, the corresponding stratified refractive-index profile can be reconstructed directly. This inverse formulation separates mathematical existence from the physical question of whether the reconstructed profile can occur in an atmosphere.

7.1 Reconstruction from a prescribed horizon law

For a monotonic profile with $n(0) = n_0$ and $n(h) > n_0$ for $h > 0$, the grazing ray derived in Sec. 2 gives the one-sided horizon distance

$$d(h) = \int_0^h \frac{n_0\, d\eta}{\sqrt{n^2(\eta) - n_0^2}}. \tag{67}$$

Because the integrand depends only on the local refractive index, differentiation of Eq. (67) with respect to its upper limit yields

$$d'(h) = \frac{n_0}{\sqrt{n^2(h) - n_0^2}}. \tag{68}$$

Solving algebraically for $n(h)$ gives the inverse reconstruction formula

$$n(h) = n_0 \sqrt{1 + \frac{1}{[d'(h)]^2}}. \tag{69}$$

Thus, within the assumptions of the model, any differentiable and strictly increasing horizon law with $d'(h) > 0$ defines a refractive-index profile. The overall multiplicative scale n_0 does not affect ray trajectories; it fixes only the absolute optical path length. Once n_0 is chosen, the profile is unique wherever $d'(h)$ is finite and nonzero.

Equation (69) also makes the near-boundary behaviour transparent. A square-root horizon law, $d(h) \sim C\sqrt{h}$, has $d'(h) \sim C/(2\sqrt{h})$ and therefore $n(h) \to n_0$ continuously as $h \to 0^+$. Expanding the square root gives

$$n(h) = n_0 \left[1 + \frac{1}{2[d'(h)]^2} + O\left(\frac{1}{[d'(h)]^4}\right)\right]. \tag{70}$$

For $d(h) \simeq \sqrt{2Rh}$, this immediately reproduces $n(h) \simeq n_0(1 + h/R)$ and hence the leading-order exponential profile found in Sec. 3.

7.2 Exact reconstruction of the spherical horizon law

We now prescribe the exact spherical arc-distance law from Eq. (53),

$$d_{\mathrm{s}}(h) = R \arccos\left(\frac{R}{R+h}\right). \tag{71}$$

Its derivative is

$$d_{\mathrm{s}}'(h) = \frac{R^2}{(R+h)\sqrt{h(2R+h)}}. \tag{72}$$

Substitution into Eq. (69) gives

$$n_{\mathrm{s}}(h) = n_0 \sqrt{1 + \frac{(R+h)^2 h(2R+h)}{R^4}}. \tag{73}$$

By construction, a grazing ray above the flat opaque boundary has exactly the same one-sided distance–height relation as a tangent line of sight above a sphere of radius R, when both models use the same numerical height h. Therefore

$$L_{\max} = d_{\mathrm{s}}(h_{\mathrm{o}}) + d_{\mathrm{s}}(h_{\mathrm{t}}) \tag{74}$$

is reproduced exactly for arbitrary observer and target heights within the idealised model.

For $h/R \ll 1$, Eq. (73) expands as

$$\frac{n_{\mathrm{s}}(h)}{n_0} = 1 + \frac{h}{R} + 2\left(\frac{h}{R}\right)^2 + O\left(\frac{h^3}{R^3}\right). \tag{75}$$

The linear term agrees with $e^{h/R} = 1 + h/R + \frac{1}{2}(h/R)^2 + \cdots$, explaining why the exponential profile reproduces the spherical law at leading order. The different quadratic terms encode the small but finite discrepancy quantified in Sec. 6.

7.3 Scope of the inverse result

The reconstruction determines the limiting grazing distance and the associated boundary-occlusion criterion. It does not by itself establish that every aspect of image formation, intensity, dispersion, turbulence, or multi-path propagation is identical to that of a spherical surface. Nor does mathematical reconstructibility imply physical plausibility. In particular, Eq. (73) increases with height, just as the exponential profile does. The next section compares this required sign and scale with those of a physical atmosphere.

8 Comparison with the Physical Atmosphere

The inverse construction of Sec. 7 shows that a stratified refractive medium can be designed mathematically to reproduce a prescribed horizon law above a flat opaque boundary. The remaining issue is whether the required profile resembles the refractive-index structure of an ordinary atmosphere. The decisive difference is the sign of the vertical refractive-index gradient.

8.1 Gradient required by the optical construction

The Earth-scale exponential model is

$$n_{\mathrm{e}}(h) = n_0 e^{h/R}, \tag{76}$$

which gives

$$\frac{dn_{\mathrm{e}}}{dh} = \frac{n_{\mathrm{e}}(h)}{R} > 0. \tag{77}$$

The exact profile reconstructed from the spherical horizon law, Eq. (73), has the same qualitative behaviour. Indeed, writing

$$\frac{n_{\mathrm{s}}^2(h)}{n_0^2} = 1 + \frac{(R+h)^2 h(2R+h)}{R^4}, \tag{78}$$

the polynomial multiplying R^{-4} is strictly increasing for $h > 0$. Therefore,

$$\frac{dn_{\mathrm{s}}}{dh} > 0 \qquad (h > 0). \tag{79}$$

Both the asymptotically matched exponential profile and the exact inverse profile thus require the refractive index to increase with altitude.

This sign also agrees with the ray geometry derived above. In a stratified isotropic medium, rays bend toward regions of larger refractive index. A profile with $dn/dh > 0$ therefore bends rays upward, allowing a ray that joins two elevated endpoints to pass through a lower turning point and graze the opaque boundary between them. This turning point is the optical analogue of a geometrical tangency point on a convex surface.

8.2 Ordinary atmospheric stratification

Except under strong local inversions, the refractive index of air decreases with altitude as density and pressure decrease [3, 11, 1, 5]. A simple idealised profile is

$$n_{\mathrm{a}}(h) \simeq 1 + (n_0 - 1)e^{-h/H}, \tag{80}$$

where H is an atmospheric scale height. Its derivative is

$$\frac{dn_{\mathrm{a}}}{dh} = -\frac{n_0 - 1}{H} e^{-h/H} < 0. \tag{81}$$

The ordinary atmosphere therefore has the opposite gradient sign to that in Eqs. (77) and (79). Rays bend toward the denser, higher-index air below rather than upward toward regions of increasing index.

This mismatch is qualitative rather than quantitative. Changing the scale height or the small surface refractivity $n_0 - 1$ modifies the magnitude of the gradient but not its sign. A monotonically decreasing profile cannot be continuously adjusted into the required increasing profile without passing through a qualitatively different stratification.

8.3 Scale of the required variation

For the exponential construction, the fractional gradient is

$$\frac{1}{n}\frac{dn}{dh} = \frac{1}{R}. \tag{82}$$

Near the boundary,

$$n(h) - n_0 \simeq n_0 \frac{h}{R}. \tag{83}$$

The exact inverse profile has the same linear term, as shown in Eq. (75). The leading requirement is therefore not an arbitrarily chosen local anomaly; it is fixed by the radius R of the spherical horizon law being reproduced.

The absolute change over terrestrial heights is small because $h/R \ll 1$. Its small magnitude, however, does not remove the sign conflict. Weak positive and negative gradients bend rays in opposite directions and therefore produce different turning-point structures.

8.4 Exceptional atmospheric conditions

Real atmospheres need not be perfectly monotonic. Temperature inversions and sharp boundary layers can produce anomalous refraction, ducting, mirages, and multiple ray paths [1, 5, 8]. Such effects may substantially shift an apparent horizon or distort individual objects. They do not, however, render the ordinary atmosphere globally equivalent to the smooth profile reconstructed above. Exact reproduction of the spherical visibility law for arbitrary endpoint heights requires the prescribed positive profile throughout the relevant height interval, rather than a local layer that occasionally bends selected rays in the required direction.

Two distinct statements should therefore be separated. First, inverse geometrical optics shows that a mathematical refractive profile above a flat boundary can reproduce the same limiting horizon law as a sphere. Second, the profile required for this equivalence is not that of an ordinary atmosphere, because its vertical gradient has the opposite sign. The construction is therefore an exact optical equivalence, but not a model of standard atmospheric refraction.

9 Discussion

The preceding results separate three questions that are often conflated in informal discussions of horizon visibility: whether refraction can generate occlusion-like ray geometry at all, whether it can reproduce the quantitative height–distance law of a sphere, and whether the required refractive profile is representative of the atmosphere. Within geometrical optics, the answer to the first two questions is affirmative. For ordinary atmospheric stratification, the answer to the third is negative.

9.1 Mathematical possibility versus physical explanation

The exponential profile provides an explicit constructive example. In the paraxial regime, it produces rays of constant upward curvature and recovers the leading spherical horizon law. Exactly, a change of variables maps its optical metric to the Euclidean exterior of a circle. Thus, as a matter of principle, optics does not forbid geometric horizon-like occlusion above a flat boundary.

This mathematical equivalence is not, by itself, a physical explanation of a terrestrial observation. Such an explanation must identify a medium with the required profile and show that the profile persists over the relevant horizontal and vertical scales. Both the exponential construction and the exact inverse reconstruction require a refractive index that increases with height, whereas the ordinary atmosphere exhibits the opposite gradient. The issue is therefore one of physical mechanism rather than fitting accuracy alone.

9.2 What the construction reproduces

The general inverse formula of Sec. 7 shows that a prescribed one-sided horizon law $d(h)$ determines a corresponding monotonic stratified profile. In particular, the exact spherical law can be reproduced for all endpoint heights within the idealised model. The idealised optical construction therefore reproduces the following properties of the limiting visibility geometry:

- a finite limiting distance for objects of fixed height;
- disappearance of lower portions before upper portions as distance increases;
- recovery of previously hidden portions when the observer is raised;
- additivity of the observer and target horizon distances.

All of these effects arise from the same grazing-ray structure that determines geometric tangency on a sphere.

The construction does not automatically reproduce every observable property of imaging above a sphere. Equality of the limiting visibility law constrains the boundary of the visible region, but does not by itself enforce equality of magnification, angular size, intensity, chromatic dispersion, or image multiplicity away from that boundary. The conformal equivalence of the exponential model is stronger than equality of a single horizon law, but even there the physical height coordinate is related nonlinearly to the radial coordinate of the circular representation. Coordinate equivalence should therefore not be confused with literal identity of the two physical systems.

9.3 Uniqueness and diagnostic value of the inverse problem

Under the assumptions of a smooth isotropic index $n(h)$ and a single grazing branch, the inverse relation

$$n(h) = n_0\sqrt{1 + \frac{1}{[d'(h)]^2}} \tag{84}$$

assigns a unique profile once n_0 is fixed. The horizon law is therefore a diagnostic constraint rather than a qualitative resemblance. A proposed optical mechanism cannot be assessed simply by noting that rays bend; its index profile must reproduce the measured dependence of limiting distance on height.

The same reasoning clarifies the limitations of ad hoc appeals to local refraction. A temperature inversion may alter a particular observation, shift an apparent horizon, or create multiple images [1, 5, 8]. Reproducing one selected ray or one selected photograph, however, does not establish a universal alternative horizon geometry. To reproduce the full family of height-dependent visibility limits, the refractive structure must satisfy the corresponding inverse profile throughout the region sampled by those rays.

9.4 Observational discrimination

At small heights, the spherical and refractive models are difficult to distinguish because their leading horizon asymptotics coincide. Meaningful discrimination must therefore rely on information beyond the leading square-root law. Useful tests include measurements over a range of observer and target heights, simultaneous meteorological profiling, repeated observations under changing atmospheric conditions, and searches for image distortion or multiple ray paths. Standard atmospheric-refraction corrections and their dependence on vertical refractivity structure provide a practical basis for such comparisons [4, 1, 5]. An atmospheric explanation predicts sensitivity to the refractive-index field, whereas the geometric model predicts stable limiting geometry once transient refractive effects are averaged or independently constrained.

The sign of the required gradient provides an especially direct diagnostic. The flat-boundary construction that mimics spherical occlusion requires upward bending rays and therefore $dn/dh > 0$ in the adopted height coordinate. A measured negative gradient does not merely weaken the proposed effect; it bends rays in the opposite direction from that required by the model.

9.5 Limitations of the analysis

The analysis is intentionally idealised. It neglects diffraction, scattering, absorption, turbulence, wavelength dependence, finite source size, and three-dimensional horizontal inhomogeneity. It also assumes a perfectly flat, opaque lower boundary and a stationary isotropic medium. These omissions do not affect the logical distinction between mathematical reconstructibility and physical plausibility, but they limit direct application to detailed atmospheric scenes.

Within these assumptions, the central conclusion is robust: refraction can be engineered to produce horizon-like occlusion above a flat surface and even to match the exact spherical height–distance law. The profile required for this purpose, however, is a specially constructed increasing-index medium rather than the ordinary atmosphere. Refraction is therefore a mathematically valid analogue of geometric occlusion, but standard atmospheric refraction is not an interchangeable explanation for it.

10 Conclusion

This work has formulated horizon occlusion as an inverse problem in geometrical optics. Starting from Fermat's principle for a horizontally stratified refractive index $n(h)$, we derived the ray equations in the paraxial and exact regimes and determined which vertical profiles reproduce the limiting visibility law associated with a curved surface.

In the paraxial approximation, constant upward ray curvature requires the exponential profile

$$n(h) = n_0 e^{h/R}, \tag{85}$$

when the curvature scale is chosen to match a sphere of radius R. The corresponding grazing rays recover the leading spherical horizon law

$$d(h) \simeq \sqrt{2Rh}, \tag{86}$$

and the maximum observer–target separation is the sum of the two one-sided horizon distances.

The exponential model is also exactly solvable. Its ray family is

$$h(x) = h_m - R\ln\left[\cos\left(\frac{x - x_m}{R}\right)\right], \tag{87}$$

and its exact one-sided horizon distance is

$$d_{\mathrm{e}}(h) = R\arccos\left(e^{-h/R}\right). \tag{88}$$

A conformal coordinate transformation maps the associated optical metric to the homogeneous Euclidean exterior of a circle. This yields an exact optical equivalence between the flat exponential medium and circular occlusion, with the important qualification that the physical height is mapped nonlinearly according to $r = Re^{h/R}$.

Comparison with the exact spherical arc-distance law,

$$d_{\mathrm{s}}(h) = R\arccos\left(\frac{R}{R+h}\right), \tag{89}$$

shows that the two models agree closely at terrestrial heights but are not identical at equal numerical height beyond leading order. Exact agreement with any prescribed horizon law can instead be imposed through the general inverse reconstruction

$$n(h) = n_0\sqrt{1 + \frac{1}{[d'(h)]^2}}, \tag{90}$$

within the assumptions of a smooth monotonic stratified medium and a single grazing branch.

The decisive physical result is the sign of the required gradient. Both the exponential construction and the profile reconstructed from the exact spherical law require

$$\frac{dn}{dh} > 0, \tag{91}$$

so that rays bend upward and possess lower turning points. Ordinary atmospheric stratification generally has the opposite sign, $dn/dh < 0$, apart from local and transient anomalous layers. Standard atmospheric refraction therefore cannot provide a global mechanism that reproduces the complete spherical horizon law above a flat boundary.

More broadly, bottom-first disappearance, recovery with observer height, and a finite horizon are not logically unique signatures of surface curvature when arbitrary refractive media are permitted. A suitably engineered optical geometry can reproduce all three. Competing explanations are distinguished not merely by the existence of bent rays, but by the quantitative refractive gradients they require and the additional imaging effects they predict. The inverse formulation thus turns a qualitative resemblance into a testable condition on the medium.

References

[1] Bradford R. Bean and E. J. Dutton. *Radio Meteorology.* Number 92 in National Bureau of Standards Monograph. U.S. Government Printing Office, Washington, D.C., 1966.

[2] Max Born and Emil Wolf. *Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light.* Cambridge University Press, Cambridge, 7 edition, 1999.

[3] Philip E. Ciddor. Refractive index of air: new equations for the visible and near infrared. *Applied Optics*, 35(9):1566–1573, 1996.

[4] Iwan Holleman and Asko Huuskonen. Analytical formulas for refraction of radiowaves from exoatmospheric sources. *Radio Science*, 48(3):226–231, 2013.

[5] International Telecommunication Union. The radio refractive index: Its formula and refractivity data. Recommendation ITU-R P.453-14, International Telecommunication Union, Geneva, 2019.

[6] W. C. Kropla and W. H. Lehn. Differential geometric approach to atmospheric refraction. *Journal of the Optical Society of America A*, 9(4):601–608, 1992.

[7] Waldemar H. Lehn and Siebren van der Werf. Atmospheric refraction: a history. *Applied Optics*, 44(27):5624–5636, 2005.

[8] B. D. Nener, N. Fowkes, and L. Borredon. Analytical models of optical refraction in the troposphere. *Journal of the Optical Society of America A*, 20(5):867–875, 2003.

[9] W. G. Rees, C. M. Roach, and C. H. F. Glover. Inversion of atmospheric refraction data. *Journal of the Optical Society of America A*, 8(2):330–338, 1991.

[10] Daniel J. Schroeder. *Astronomical Optics*. Academic Press, San Diego, 2 edition, 2000.

[11] U.S. Committee on Extension to the Standard Atmosphere. U.S. Standard Atmosphere, 1976. Technical Report NASA-TM-X-74335, U.S. Government Printing Office, Washington, D.C., 1976. NASA report identifier 19770009539.